CAFE-Net: Context-Aware Feature Enhancement for Reliable Multiscale Detection of Martian Craters

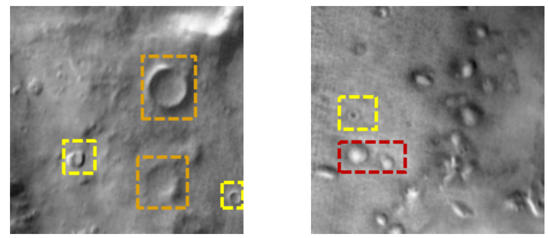
Zhuohang Li, Xinyao Wang, Zhaozhao Zhang, Tongtong Shen, and Wen Zheng

***Abstract*—Reliable recognition of Martian impact craters is critical for spacecraft landing site selection and future surface exploration. However, this task remains challenging due to the pronounced multiscale characteristics of craters and interference from crater-like structures such as Martian domes. Notably, the ejecta textures surrounding craters provide important discriminative cues for distinguishing craters from domes. To address these challenges, a context-aware and feature-enhanced net (CAFE-Net) is proposed for accurate Martian crater detection in complex environments. CAFE-Net incorporates three key components: a multiscale context-aware (MSCA) module to capture local texture information, a feature enhancement unit (FEU) to alleviate the loss of small-scale features, and a channel self-learning fusion strategy to strengthen feature representation via channelwise interactions. Extensive experiments demonstrate that the proposed method achieves improvements of 3.94% in precision and 3.13% in recall, with an F_1-score of 86.41%, outperforming existing methods while maintaining efficient inference. However, the proposed method is currently limited in handling nighttime imagery and severely weathered craters. Future work will explore multisource data fusion to improve robustness and support applications such as vision-based landing site selection for Mars missions.**

***Index Terms*—Feature fusion, local context information, object detection, one-stage object detection, reliable Martian craters detection.**

I. Introduction

MARS exhibits highly complex surface structures. Analysis of remote sensing imagery acquired from orbital satellites indicates that, unlike the Moon and other planetary bodies, Martian impact craters present pronounced multiscale distribution characteristics, with diameters ranging approximately from 10 to 100 pixels, as illustrated in Fig. 1(a). In addition, a large number of Martian domes are widely distributed around impact craters. These domes typically appear as dome-shaped hills, such as Aonia Tholus (54 km in diameter, 59.0°S, 280.0°E) and E. Mareotis Tholus (5.0 km in diameter, 36.24°N, 274.87°E), as shown in Fig. 1(b). Under nadir-view observation conditions, these dome structures exhibit enhanced central brightness but lack the rich textural details commonly observed in impact craters [1].



(a) (b)

Fig. 1. (a) Illustrates impact craters of varying radii. (b) Yellow bounding boxes indicate craters, while red bounding boxes highlight domes. A distinguishing characteristic of domes is their enhanced central brightness in imagery.

Traditional crater detection approaches mainly rely on manual annotation, which is labor-intensive and time-consuming, while also suffering from limited accuracy and frequent missed and false detections [2]. To improve detection efficiency, machine learning-based methods employing edge extraction and shape matching have been proposed. However, due to the high geometric similarity between craters and domes in remote sensing imagery, shape-based methods demonstrate limited discriminative capability, making reliable crater identification challenging [3] [4].

In recent years, the remarkable success of deep learning in generic visual feature extraction has promoted its application to automated crater detection [5] [6] [7]. Among these approaches, the YOLO family has attracted considerable attention due to its high efficiency [8]. Some studies attempt to improve detection accuracy by modeling the spatial distribution characteristics of craters. For example, Mu et al. [9] introduce the CBAM into YOLO and propose the YOLO-Crater model, which captures crater distribution features by modeling spatial and channel dependencies. Yu et al. [10] enhance long-range dependence modeling in YOLOv9.

Received 16 January 2026; revised 27 March 2026; accepted 4 May 2026. Date of publication 8 May 2026; date of current version 26 May 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 11702289, in part by the One Hundred Talent Project of the Chinese Academy of Sciences under Grant Y5ZKRCGZ01, and in part by the Science and Technology Commission of Shanghai through the Second Batch of the 2024 Annual Science and Technology Innovation Action Plan in the Field of Aerospace and Marine Science and Technology under Grant 24DZ3101402. *(Corresponding author: Xinyao Wang.)*

Zhuohang Li, Zhaozhao Zhang, and Wen Zheng are with the Key Laboratory for Satellite Digitalization Technology, Innovation Academy for Microsatellites, Chinese Academy of Sciences, Shanghai 201304, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: lizhuohang2024@mail.ustc.edu.cn; zhangzhaozhao24@mails.ucas.ac.cn; zhengwen@microsate.ac.cn).

Xinyao Wang is with the Key Laboratory for Satellite Digitalization Technology, Innovation Academy for Microsatellites, Chinese Academy of Sciences, Shanghai 201304, China (e-mail: wangxinyao@microsate.ac.cn).

Tongtong Shen is with the College of Artificial Intelligence, Taiyuan University of Technology, Jinzhong, Shanxi 030600, China (e-mail: 2024319017@link.tyut.edu.cn).

Digital Object Identifier 10.1109/LGRS.2026.3691336

Although these methods partially consider distribution characteristics, they do not explicitly model crater-like interference objects, such as Martian domes, resulting in limited discrimination capability and relatively high false detection rates. To further improve recognition reliability, Guo et al. [11] proposed Crater-DETR, which incorporates Transformer-based attention mechanisms to model global dependencies. While achieving improved detection accuracy, this approach incurs substantial computational overhead, limiting its applicability on resource-constrained onboard satellite platforms. Consequently, achieving a favorable balance between reliable recognition and inference efficiency remains a critical challenge in Martian crater detection.

To address these challenges, this work focuses on two key aspects: mitigating the degradation of small-scale feature representations caused by increasing network depth and enhancing the modeling of intrinsic crater characteristics, particularly local texture information. Based on these considerations, a reliable Martian crater detection framework, termed context-aware and feature-enhanced net (CAFE-Net), is developed. The main contributions of this work are summarized as follows.

1) A multiscale context-aware (MSCA) module is introduced to capture local contextual textures surrounding craters. By exploiting the richer contextual patterns of craters compared to domes, the MSCA module enhances crater-dome discrimination and improves detection accuracy.
2) A feature enhancement unit (FEU) is embedded before the feature fusion network to mitigate the loss of small-scale feature representation in the backbone. In addition, a channel-level self-learning fusion strategy is adopted to assign adaptive weights to different channels, leveraging channel diversity to strengthen crater feature representation in feature maps and improve crater recognition across different terrains and scales.

II. Methodology

YOLO is a widely used one-stage object detection framework that employs the C2f module and PANet-based feature aggregation to achieve high detection efficiency. Although it performs well on general-purpose detection tasks, its effectiveness degrades when applied to Martian crater detection due to the complex and heterogeneous surface topography. To address this limitation, an improved model, termed CAFE-Net, is developed based on the YOLOv8 architecture, as shown in Fig. 2. CAFE-Net incorporates three key enhancements: 1) an MSCA module for capturing crater texture information across different scales; 2) an FEU to improve small-scale feature representation in the backbone; and 3) a high-efficiency feature fusion network (HEFFN) that strengthens small-scale feature learning through channelwise information exploitation. Detailed descriptions of these components are provided in Sections II-A–II-C.

A. Mutliscale Context-Aware Module

In satellite remote sensing imagery, due to the commonly adopted top-down imaging geometry, dome structures and

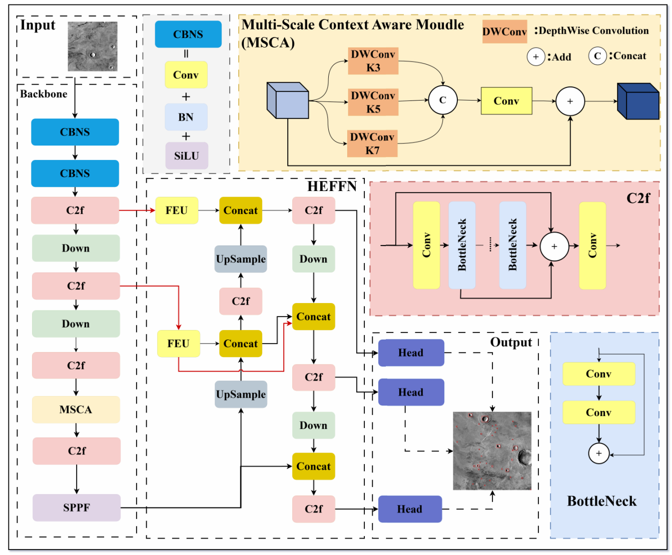


Fig. 2. Overall architecture of CAFE-Net and MSCA modules.

impact craters exhibit a high degree of visual similarity, typically appearing as approximately elliptical objects. Meanwhile, the discriminative background information surrounding the targets is easily degraded by image noise, which constrains the effective extraction of discriminative features in the crater neighborhood. As a result, misclassification between craters and domes remains a persistent challenge.

Recent studies have attempted to distinguish craters from domes by modeling temporal and spatial dependencies within feature maps [12]; however, the overall performance improvement remains limited. Notably, although the two categories share strong geometric similarity, impact craters usually contain richer local texture information in their surrounding regions compared to domes. Motivated by this multiscale contextual discrepancy inherent in remote sensing imagery, an MSCA module is introduced (see Fig. 2), employing parallel depthwise separable convolutions with kernel sizes of 3, 5, and 7 to capture multiscale crater textures. The fused features improve discrimination between craters and domes while reducing computational complexity and accelerating inference.

B. Feature Enhancement Unit

Due to the varying sizes of craters distributed across the Martian surface and the inherent limitations of existing deep learning-based backbone networks in extracting discriminative features, an FEU is proposed to improve the representation of small-scale craters in complex remote sensing imagery. The FEU is designed with a multibranch architecture incorporating different convolutional kernel sizes to enhance feature diversity and multiscale perception. As illustrated in Fig. 3, it consists of three branches: the first branch, inspired by residual structures, preserves critical information from the original input while mitigating gradient explosion during training by generating an equivalent mapping. The other two branches apply cascaded standard convolutions with kernel sizes of 1×3 and 3×1, respectively, to capture varying sizes of craters more effectively. The mathematical expressions of FEU can be written as follows:

$$W_1 = f_{\text{conv}}^{3\times3}\left\{f_{\text{conv}}^{3\times1}\left[f_{\text{conv}}^{1\times3}\left(f_{\text{conv}}^{1\times1}(F)\right)\right]\right\} \tag{1}$$
$$W_2 = f_{\text{conv}}^{3\times3}\left\{f_{\text{conv}}^{1\times3}\left[f_{\text{conv}}^{3\times1}\left(f_{\text{conv}}^{1\times1}(F)\right)\right]\right\} \tag{2}$$
$$Y_1 = \text{Concat}(W_1, W_2) \tag{3}$$
$$Y_2 = f_{\text{conv}}^{1\times1}(F) \tag{4}$$
$$Out = Y_1 \oplus Y_2 \tag{5}$$

where $f_{\text{conv}}^{3\times3}$, $f_{\text{conv}}^{3\times1}$, $f_{\text{conv}}^{1\times3}$, and $f_{\text{conv}}^{1\times1}$ represent the standard convolution operations with different kernel sizes. Concat$(\cdot)$ is the feature map concatenation operation, and $\oplus$ represents the elementwise addition operation of the feature map. F is the input feature map, W_1 and W_2 represent the intermediate output features after a series of convolutions, Y_1 denotes the concatenated feature map W_1 and W_2, Y_2 is the feature map obtained by applying a 1×1 convolution to F, and the Out represents the output feature map FEU.

By embedding the FEU into the neck routing stage of the network, the overall capacity to extract and distinguish crater features of varying scales in complex imagery is further enhanced.

C. High Effect Feature Fusion Net

Although the MSCA module enhances local contextual modeling and improves the discrimination between domes and craters, small-scale crater recognition remains challenging due to insufficient fusion of low-level spatial details and high-level semantic information. In scenes containing craters of varying sizes, small targets are, therefore, prone to being overlooked. To address this limitation, Tan et al. [13] proposed the bidirectional feature pyramid network (BiFPN), which exploits bidirectional information flow and learnable feature fusion for effective multiscale representation. Inspired by this design, an improved architecture termed the high effect feature fusion network is introduced, as illustrated in Fig. 3.

Unlike BiFPN that typically applies uniform fusion weights across entire feature maps, the CRC strategy first concatenates the feature maps and then multiplies the normalized trainable weights with the same number of channels, as shown in Fig. 4, and the formula expressions of CRC can be written as follows:

$$Out = \sum_u \frac{w_u}{\varepsilon + \sum_m w_m} \cdot x_u \tag{6}$$

where w_u represents the trainable weight in the uth channel, m represents the total number of channels after concatenation, and ε is set to 0.0001 to avoid numerical instability.

This enables discriminative channel-level weighting, improving the representation of small-scale targets. As a result, the CRC strategy further strengthens multiscale semantic fusion, thereby enabling better performance than the original model in multiscale crater detection tasks by providing more informative features for subsequent crater detection and recognition.

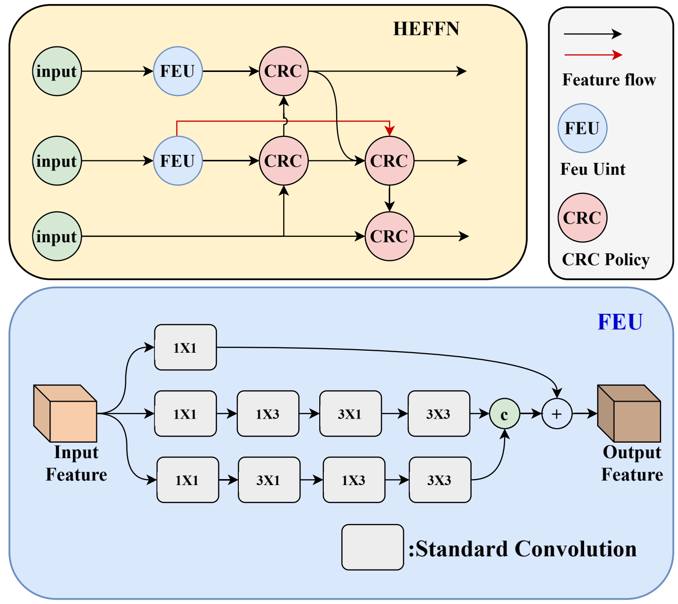


Fig. 3. Structure of the HEFFN and FEU.

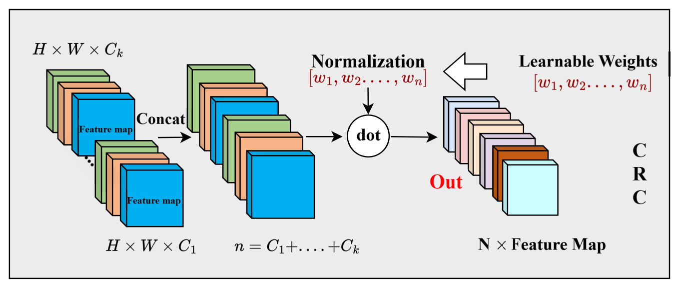


Fig. 4. Method of CRC strategy.

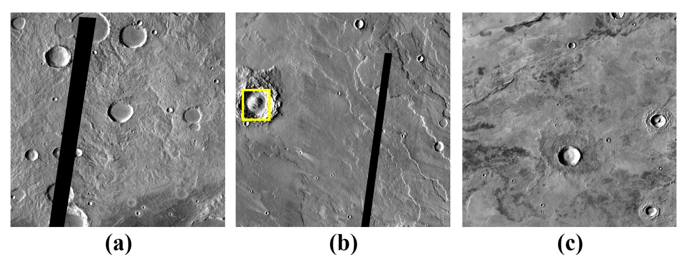


Fig. 5. Examples of MDCD images requiring correction (a) images with black borders introduced by cropping that distort crater structures and should be removed, (b) images requiring reannotation of volcanic domes, indicated by yellow bounding boxes, and (c) valid images requiring neither removal nor reannotation.

III. Experiments and Results

A. Datasets and Environment Description

Three Martian crater datasets are used in this study: the MDCD dataset [14], the DACD dataset [15], and a subset of global Martian images from the 2022 GeoAI Martian Challenge on CodaLab. All datasets consist of daytime remote sensing images under normal illumination. MDCD images with black bands caused by improper cropping are removed. In addition, since craters are characterized by surrounding ejecta textures, annotations of part of the MDCD dataset are

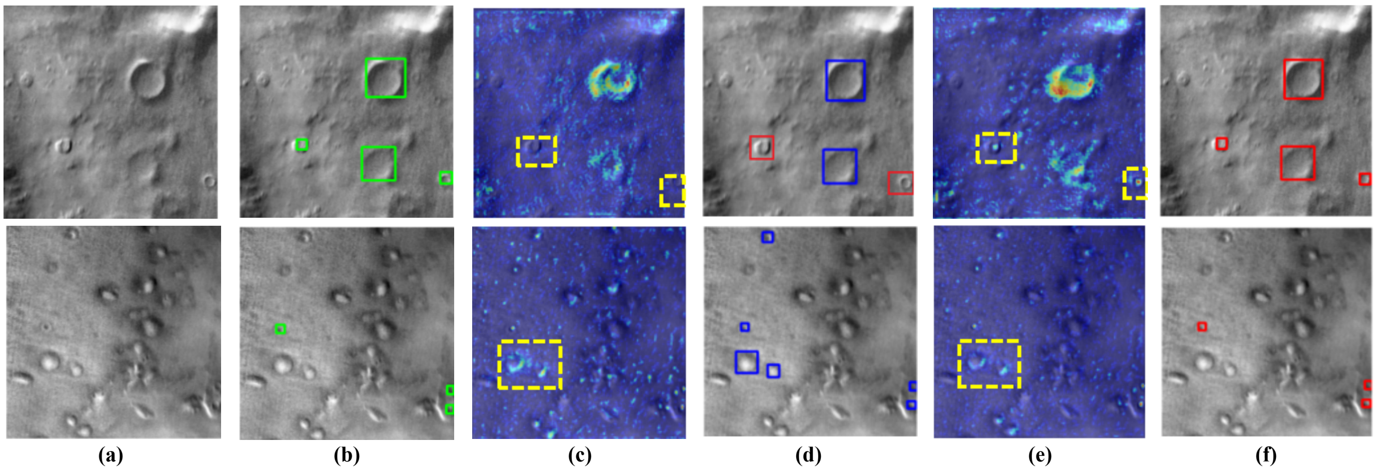


Fig. 6. Illustration of detection performance in small-scale crater detection and comparison on distinguishing craters from volcanic domes. The first row illustrates the small-scale crater detection results, while the second row displays the comparative outcomes of distinguishing craters from volcanic domes. In detail, (a) original remote sensing image. (b) Ground-truth annotation. (c) Heatmap generated by the baseline model. (d) Prediction result of the baseline model, where blue boxes indicate the model predictions. (e) Heatmap generated by the proposed model. (f) Prediction result of the proposed model.

TABLE I
COMPARISON RESULTS

No.	Model Method	Precision	Recall	F_1 Score	OA	mAP_{50}	$mAP_{50:95}$	Inference Time (s)
1	YOLOv8 [16]	83.76 ± 0.32	82.61 ± 0.41	83.18 ± 0.29	85.24 ± 0.27	87.92 ± 0.36	55.84 ± 0.44	0.021
2	Fast-RCNN [17]	83.34 ± 0.45	80.34 ± 0.38	81.81 ± 0.34	83.72 ± 0.30	86.25 ± 0.51	54.18 ± 0.47	0.162
3	TridentNet(2022) [18]	84.26 ± 0.37	82.65 ± 0.40	83.45 ± 0.31	85.51 ± 0.28	87.98 ± 0.42	55.76 ± 0.39	0.094
4	YOLO-Crater(2023) [9]	84.36 ± 0.29	82.76 ± 0.33	83.55 ± 0.28	85.64 ± 0.26	88.41 ± 0.35	56.37 ± 0.41	0.024
5	YOLO-ADCN(2024) [10]	85.42 ± 0.41	80.85 ± 0.46	83.07 ± 0.39	85.02 ± 0.29	88.96 ± 0.44	56.48 ± 0.38	0.028
6	YOLOv11(2024) [19]	84.53 ± 0.36	82.85 ± 0.34	83.68 ± 0.30	85.79 ± 0.25	88.74 ± 0.37	56.92 ± 0.35	0.026
7	Crater-DETR(2024) [11]	84.20 ± 0.48	81.64 ± 0.52	82.90 ± 0.46	84.87 ± 0.31	87.21 ± 0.49	55.02 ± 0.43	0.138
8	YOLOv13(2025) [20]	83.65 ± 0.44	80.17 ± 0.39	81.88 ± 0.36	83.80 ± 0.29	87.36 ± 0.41	54.92 ± 0.45	0.031
9	**CAFE-Net**	**87.10 ± 0.21**	**85.74 ± 0.18**	**86.41 ± 0.16**	**88.57 ± 0.19**	**90.83 ± 0.19**	**59.64 ± 0.22**	**0.019**

manually refined, as shown in Fig. 5. The GeoAI covers global crater-dome scenes, while MDCD and DACD provide regional multiscale distributions and complex local scenarios, respectively, enabling evaluation across varied Martian environments. The final dataset contains 2000 images (256×256 pixels) with approximately 30 000 craters, split into 1000 training, 500 validation, and 500 test samples.

All models are implemented using PyTorch and trained for 300 epochs with the Adam optimizer. The initial learning rate is set to 0.001 with a cosine annealing schedule. Geometric transformations are applied for data augmentation. Experiments are conducted on an NVIDIA RTX 3090 GPU (24 GB), with a batch size of 8.

B. Evaluation Metrics

In this experiment, precision (P), recall (R), $F1$-score, and mean average precision (mAP) are adopted as the primary evaluation metrics. Precision and recall are computed based on the numbers of true positives (TPs), false positives (FPs), and false negatives (FNs). An intersection over union (IoU) threshold of 0.75 is employed to distinguish TP and FP, where TP denotes correctly detected craters, FN represents missed craters, and FP indicates falsely detected craters. The mAP metric is reported in terms of mAP_{50} and $\text{mAP}_{50:95}$, which reflect detection performance under different IoU thresholds. The $F1$-score provides a balanced measure of precision and recall, offering a comprehensive evaluation of the model's recognition performance. Overall accuracy (OA) is introduced to assess overall prediction correctness, representing the proportion of correctly detected targets among all detection outcomes, computed based on TP, FP, and FN. The corresponding formulations are defined as follows:

$$F_1 = \frac{2 \times P \times R}{P + R}, \quad OA = \frac{TP}{TP + FP + FN}. \tag{7}$$

C. Experimental Results

The proposed CAFE-Net is comprehensively compared with several state-of-the-art crater detection models commonly adopted in recent years. As reported in Table I, all evaluation results are presented as mean $\pm$ standard deviation (std) over multiple independent runs; compared with the baseline YOLOv8, CAFE-Net achieves improvements of 3.34% in precision and 3.23% in F_1 score. In addition, when compared with YOLOv9-ADCN, a notable gain of 4.89% is observed in recall. It is worth emphasizing that even relative to the most recent YOLOv13, the proposed method consistently demonstrates superior performance in terms of precision, recall, and F_1 score. Table I further presents a comparison

TABLE II
ABLATION STUDY

FEU	HEFFN	MSCA	P	R	F_1	mAP_{50}	$mAP_{50:95}$
×	×	×	83.76	82.61	83.18	87.92	55.84
✓	×	×	84.30	83.37	83.83	88.21	56.10
×	✓	×	84.12	82.95	83.53	88.21	56.08
×	×	✓	84.58	83.10	83.83	88.67	56.54
✓	✓	×	85.02	83.84	84.42	89.12	57.10
×	✓	✓	85.21	83.42	84.30	89.38	57.42
✓	✓	✓	**87.10**	**85.74**	**86.41**	**90.83**	**59.64**

of inference time among different models. Compared with two-stage detectors and the Transformer-based Crater-DETR, CAFE-Net requires significantly less inference time, and it also exhibits higher inference efficiency than other YOLO-based methods.

To validate the effectiveness of the core components in CAFE-Net, ablation experiments are conducted, with results reported in Table II. Embedding the FEU module and CRC fusion strategy into the HEFFN network alleviates small-scale feature loss in the deep backbone and assigns adaptive channel weights, strengthening small-scale crater representation and increasing precision from 83.76% to 85.02%. Further incorporation of the MSCA module captures local texture patterns surrounding craters, reducing misclassification between craters and visually similar landforms, thereby improving precision to 87.10% and F_1 score to 86.41%.

Qualitative results are illustrated in Fig. 6. In the first row, Martian scenes containing craters of multiple scales are presented. The heatmap comparison indicates that the proposed model allocates greater attention to small-scale craters than the baseline method, effectively reducing missed detections of small targets. The second row depicts scenarios where domes and craters coexist. In such complex terrains, the proposed model exhibits reduced response to dome-like structures, thereby alleviating confusion between domes and craters and improving overall detection accuracy.

IV. CONCLUSION

This letter proposes a Martian crater detection model termed CAFE-Net, which improves the detection performance of small-scale craters in daytime Martian remote sensing imagery while enhancing the discrimination between domes and craters. By incorporating an MSCA module, the proposed model captures local texture patterns around craters, thereby strengthening their feature representation relative to domes. Moreover, a CRC fusion strategy and an FEU are embedded into the feature fusion network, effectively alleviating the limitations of existing deep learning methods in small-scale feature extraction and channel utilization. Experimental results on the constructed test dataset demonstrate that CAFE-Net consistently outperforms representative models, including the Transformer-based Crater-DETR [11] (2024) and YOLOv13 [20] (2025), across all evaluation metrics, achieving a precision of 87.10%, a recall of 85.74%, and an F_1 score of 86.41%. Nevertheless, the proposed approach does not yet fully address challenging scenarios, including nighttime Martian imagery and severely weathered crater recognition, and model robustness under varying illumination conditions and degraded terrains warrants further investigation. Future work will investigate multisource data fusion, such as integrating multispectral imagery with DEMs derived from LiDAR measurements, to address nighttime crater detection and the recognition of weathered craters, thereby improving the stability and generalization of the proposed model and supporting its potential application to vision-based landing site selection for future Mars missions.

REFERENCES

[1] W. Cao, Z. Xiao, R. Xu, and Y. Wang, "Extracting background secondary Craters based on fusion of multiscale and multifacies Crater topography information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4511516.

[2] Y. Tao, S. H. G. Walter, J.-P. Müller, Y. Luo, and S. Xiong, "A high-resolution digital terrain model mosaic of the Mars 2020 perseverance rover landing site at Jezero Crater," *Earth Space Sci.*, vol. 10, no. 10, pp. 1–16, 2023.

[3] Q. Duan, R. Liu, and Y. Liu, "Automatic lunar Crater detection based on dem data using a max curvature detection method," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[4] J. S. Marques and P. Pina, "Crater delineation by dynamic programming," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1581–1585, Jul. 2015.

[5] X. Kang, H. Yin, and P. Duan, "Global-local feature fusion network for visible-infrared vehicle detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[6] Z. Li, W. Liu, Z. Xie, X. Kang, P. Duan, and S. Li, "FAA-Det: Feature augmentation and alignment for anchor-free oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3504598.

[7] X. Zheng, J. Bi, K. Li, G. Zhang, and P. Jiang, "SMN-YOLO: Lightweight YOLOv8-based model for small object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[8] M. Hussain, "YOLOv1 to v8: Unveiling each variant—A comprehensive review of YOLO," *IEEE Access*, vol. 12, pp. 42816–42833, 2024.

[9] L. Mu, L. Xian, L. Li, G. Liu, M. Chen, and W. Zhang, "YOLO-Crater model for small Crater detection," *Remote Sens.*, vol. 15, no. 20, p. 5040, Oct. 2023.

[10] Z. Yu, S. Fong, and R. C. Millham, "Small-scale Martian Crater detection by deep learning with enhanced capture of features information and long-range dependencies," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[11] Y. Guo, H. Wu, S. Yang, and Z. Cai, "Crater-DETR: A novel transformer network for Crater detection based on dense supervision and multiscale fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 3376398.

[12] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Computer Vision—ECCV 2018*, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds., Cham, Switzerland: Springer, Oct. 2018, pp. 3–19.

[13] M. Tan, R. Pang, and Q. V. Le, "EfficientDet: Scalable and efficient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10778–10787.

[14] S. Yang and Z. Cai, "High-resolution feature pyramid network for automatic Crater detection on Mars," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4601012.

[15] S. Yang and Z. Cai, "Progressive domain adaptive network for Crater detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 3215741.

[16] R. Varghese and S. Sambath, "YOLOv8: A novel object detection algorithm with enhanced performance and robustness," in *Proc. Int. Conf. Adv. Data Eng. Intell. Comput. Syst. (ADICS)*, 2024, pp. 1–6.

[17] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," 2015, *arXiv:1506.01497*.

[18] D. Paz, H. Zhang, and H. I. Christensen, "Tridentnet: A conditional generative model for dynamic trajectory generation," in *Intelligent Autonomous Systems*. Cham, Switzerland: Springer, 2022, pp. 403–416.

[19] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

[20] M. Lei et al., "YOLOv13: Real-time object detection with hypergraph-enhanced adaptive visual perception," 2025, *arXiv:2506.17733*.